

3. PAIR OF LINEAR EQUATION IN TWO VARIABLES

Marks: 20

MCQ's TEST – 2

Time: 30 min



1. Sum of two numbers is 50 and their difference is 10, then the numbers are
(a) 30 and 20 (b) 24 and 14 (c) 12 and 2 (d) none of these
2. The sum of the digits of a two-digit number is 12. The number obtained by interchanging its digits exceeds the given number by 18, then the number is
(a) 72 (b) 75 (c) 57 (d) none of these
3. The sum of a two-digit number and the number obtained by interchanging its digits is 99. If the digits differ by 3, then the number is
(a) 36 (b) 33 (c) 66 (d) none of these
4. Seven times a two-digit number is equal to four times the number obtained by reversing the order of its digits. If the difference between the digits is 3, then the number is
(a) 36 (b) 33 (c) 66 (d) none of these
5. A two-digit number is 4 more than 6 times the sum of its digits. If 18 is subtracted from the number, the digits are reversed, then the number is
(a) 36 (b) 46 (c) 64 (d) none of these
6. The sum of two numbers is 1000 and the difference between their squares is 25600, then the numbers are
(a) 616 and 384 (b) 628 and 372 (c) 564 and 436 (d) none of these
7. Five years ago, A was thrice as old as B and ten years later A shall be twice as old as B, then the present age of A is
(a) 20 (b) 50 (c) 30 (d) none of these
8. The sum of thrice the first and the second is 142 and four times the first exceeds the second by 138, then the numbers are
(a) 40 and 20 (b) 40 and 22 (c) 12 and 22 (d) none of these
9. The sum of twice the first and thrice the second is 92 and four times the first exceeds seven times the second by 2, then the numbers are
(a) 25 and 20 (b) 25 and 14 (c) 14 and 22 (d) none of these
10. The difference between two numbers is 14 and the difference between their squares is 448, then the numbers are
(a) 25 and 9 (b) 22 and 9 (c) 23 and 9 (d) none of these

11. The solution of the system of linear equations $\frac{x}{a} + \frac{y}{b} = a + b$, $\frac{x}{a^2} + \frac{y}{b^2} = 2$ are
 (a) $x = a$ and $y = b$ (b) $x = a^2$ and $y = b^2$ (c) $x = 1$ and $y = 1$ (d) none of these
12. The solution of the system of linear equations $2(ax - by) + (a + 4b) = 0$, $2(bx + ay) + (b - 4a) = 0$ are
 (a) $x = a$ and $y = b$ (b) $x = -1$ and $y = -1$ (c) $x = 1$ and $y = 1$ (d) none of these
13. If $31x + 43y = 117$ and $43x + 31y = 105$, then the value of $x - y$ is:
 (a) $1/3$ (b) -3 (c) 3 (d) $-1/3$
14. If $\frac{x}{2} + y = 0.8$ and $\frac{7}{\left(x + \frac{y}{2}\right)} = 10$ then the value of $x + y = \dots$
 (a) 1 (b) -0.8 (c) 0.6 (d) 0.5
15. If $(6, k)$ is a solution of the equation $3x + y - 22 = 0$, then the value of k is:
 (a) 4 (b) -4 (c) 3 (d) -3
16. If $\sqrt{a}x - \sqrt{b}y = b - a$ and $\sqrt{b}x - \sqrt{a}y = 0$ then the value of x, y is
 (a) $a + b$ (b) $a - b$ (c) $\sqrt{ab}$ (d) $-\sqrt{ab}$
17. The equations $x - y = 0.9$, $\frac{11}{x + y} = 2$ have the solution
 (a) $x = 5, y = a$ (b) $x = 3.2, y = 2.3$ (c) $x = 3, y = 2$ (d) none of these
18. If $\frac{2}{x} - \frac{3}{y} = 13$ and $\frac{5}{x} - \frac{4}{y} = -2$ then $x + y$ is....
 (a) $-1/6$ (b) $1/6$ (c) $5/6$ (d) $-5/6$
19. If $3x - 5y = 1$, $\frac{2x}{x - y} = 4$ then the value of $x + y$ is
 (a) $\frac{1}{3}$ (b) 3 (c) -3 (d) $-\frac{1}{3}$
20. If $2^x \cdot 8^y = 64$ and $4^x \cdot 2^y = 32$ then x, y is.....
 (a) $3, 1$ (b) $1, 3$ (c) $2, 2$ (d) None of these